

Initialisation

Split months by volatility $\rightarrow$ starting guesses for $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}$

Repeat for $M = 2,000$ iterations (first $B = 500$ discarded as burn-in)

Step 1: Sample the hidden state path (FFBS)

Holds $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}$ fixed

Step 2: Update the emission parameters (NIW posterior)

Holds $\{s_t\}$ and $\mathbf{P}$ fixed

Step 3: Update the transition matrix (Dirichlet posterior)

Holds $\{s_t\}, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ fixed

Regime identification

Label switching: state with higher $\bar{\mu}_{\text{VOL}} \rightarrow$ panic

Output

Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k, \bar{\boldsymbol{\Sigma}}_k, \bar{\mathbf{P}}$

Apply forward filter with posterior means $\rightarrow \pi_t^{\text{filter}}$